

Megadrought Notes

Seven years later

Wolkovich & Donahue

Last updated: February 2026 (Kan'ehoe)

Contents

1	In later months of 2026...	1
2	Returning from 2019 ...	2
3	What happened before?	2
4	Figures	4

1 In later months of 2026...

5 October 2026 discussion

Victor, Megan, Lizzie: Some notes I made directly in the megadroughts.tex and on GitHub issues.

We discussed a few things that led us to think we should do some lit review ... I will start a git issue. See new issue 76.

We review issue 74 (Potential underlying models of native species germination) and discussed how Case 3 could potentially enhance buffered population growth ... but may depend on mortality assumptions for those seeds.

We need to find some lit on how life history strategy affects speed of adaptation.

28 September 2026 discussion

Victor, Megan, Lizzie: Some notes I made directly in the megadroughts.tex.

Bet-hedging: We have bet-hedging (natives are betting a better year is coming!), but we should not think of the species as bet-hedging. That's BAD! Stop doing that Lizzie.

Bet-hedging would be the evolutionary explanation for why these two strategies would work.

Usual lit on bet-hedging: I have a bunch of babies, should I set them up to (for example) disperse far or close? There is usually an individual that can ration a set of resources. In contrast our seeds are all the same (and they have no parental investment).

10 September 2026 Victor, Lizzie, Megan ...

... see also notes from some summer meetings on wiki and in issues on GitHub

- Megan asks: Is tracking the really important (she then talked herself into this: yes, it expands environments under which high germination occurs)
- ... but then tracking terminology is not helpful
- should we replace alpha with mean germination?
- rename alpha: germination plasticity? Germinates under a wider range of climates (maybe?)
- seedbank survival is not germinating (good survivors are poor germinators) and overwintering (another component)

Paper angle:

- germination and survival (inversely related)
- invasive species are successful because they germinate under a wide range of conditions. Are there conditions under which that strategy becomes a detriment to its success with climate change?
 - extended periods where high germination leads to germination failure (seedbank failure) which is either persistent drought or long-mean shifts?
 - colors of noise?

To do next!

- Frame the paper! Then work on the plots.
- Lizzie will draft an abstract and outline some points.

2 Returning from 2019 ...

Okay! We're back. We're back to megadroughts. Yesterday, Megan thought hard and made some really good notes issue # 66. And I spent a while trying to recreate what the hell we did (see section 'What happened before?').

I also spent a smidge of time trying to find out if anyone ever used the Ellner, Snyder *et al.* paper 'An expanded modern coexistence theory for empirical applications' which:

... present[s] a general computational approach that extends our previous solution for the storage effect to all of standard MCT's spatial and temporal coexistence mechanisms, and also process-defined mechanisms amenable to direct study such as resource partitioning, indirect competition, and life history trade-offs.

It's been cited a good bit, but after looking at the top 10-ish articles, I could not find one that actually used it. Has **anyone ever used it?** or do they all just talk about it (especially the Levine group)? For example, Detto, Levine, Pacala's 2022 'Maintenance of high diversity in mechanistic forest dynamics models of competition for light' says:

Recent studies by (Ellner et al., 2016, 2019) provide the needed simulation tools for modern coexistence theory to test the conjectures we pose here.

3 What happened before?

17 February 2026

Basically, it seems like we were advancing on both the tracking paper and the megadroughts, especially in late 2018 and through 2019. We had model runs etc.. Then in May 2019, Megan pushes more runs and I appear to go mute (see especially issue 39. Sometime after this we're pretty sure we focus all efforts on tracking and drop megadroughts (I mean, Covid hit in March 2020); I even recall a call where Megan asked if she could delete out the megadroughts flags in the model run scripts and I said something like 'oh sure.'

When we stopped we were struggling with getting enough single species to survive (and also co-existing species, of course).

Possibly useful links and timeline info from my poking around the repo in February 2026.

1. Issue 39, which is called check the megadrought runs!
2. `docs/notes/megadroughts.txt` is notes I set up in 2018 and was updating that year.
3. `docs/manuscripts/megadroughts.tex` is a file I set up in 2018 from the main `VarEnv_notes` file (now only on master) when I was trying to pull out all the relevant megadrought info.... this is also mentioned in issue 15
4. In August 2018, I updated about concerns we had, "Based on our current model, we're not entirely convinced that drought could drive the invaders extinct via a wider (potentially shorter) germination curve. This is because the exotics would still go in many years (and potentially more years than the natives given that they have narrower curves, depending on the parameterization) and even a small amount of biomass could be converted to seed, thus the exotics would not go extinct."
5. Not sure this is helpful, but here are all the commits. I looked at some. They jive with what I write here.
6. Other issues with megadrought in them are not so helpful

(?)

4 Figures



Figure 1: I am a caption.